MedSkill-E^2: Evaluating and Evolving Medical Agent Skills at Scale

Anonymous ACL submission

Abstract

LLM agents increasingly depend on external skills to perform medical tasks, yet existing benchmarks measure only end-to-end task success—not whether the agent correctly executes the skill specification it was given. We introduce MED-SKILLBENCH, the first large-scale skill-level execution benchmark for medical AI agents, comprising 1,462 skills from three open-source repositories evaluated in isolated sandboxes with a mandatory read-first protocol and eight-dimensional binary scoring. Our evaluation reveals a systematic *read-but-not-use* (RBU) gap: agents invoke skills at 96.2% but complete specified tasks at only 74.3%, a 22-percentage-point specification grounding failure. Stratifying by skill interface type shows that guide-style skills (no executable tool) exhibit the largest gap (0.269), while tool-based skills show substantially smaller gaps (0.069–0.111), identifying description groundability—not environment dependency—as the primary bottleneck. To close this gap, we propose GEPA-MED, a reflective prompt evolution method that uses execution trajectories and judge feedback to iteratively rewrite skill descriptions without model weight changes. MEDSKILL-E^2 establishes a framework that shifts medical agent skills from static documentation to evolvable, evaluation-driven interfaces.

1 Introduction

Large language model (LLM) agents increasingly rely on external *skills*—structured tool descriptions that extend model capabilities beyond parametric knowledge—to perform domain-specific tasks. Recent work demonstrates that curated skill libraries substantially improve agent performance: **?** report an average gain of 16.2 percentage points across domains, with the medical domain exhibiting the largest improvement at 51.9 percentage points. This finding underscores the critical role of skill quality in high-stakes settings such as clinical decision support, biomedical literature analysis, and drug safety assessment.

Despite this promise, three gaps hinder progress toward reliable medical agent skills.

Gap 1: Task-level evaluation obscures skill-level failures. Existing medical agent benchmarks evaluate end-to-end task success (**???**) but do not isolate whether the agent correctly *executed* the skill specification it was given. An agent may answer a question correctly by ignoring a poorly written skill description entirely, or it may fail a task despite faithfully following a description that lacks critical information. Task-level metrics conflate these fundamentally different failure modes.

Gap 2: Skills are treated as static artifacts. Current skill development follows a "write once, deploy forever" paradigm (**??**). Once a `SKILL.md` file is authored, it is rarely revisited—even when downstream execution data reveals systematic failure patterns. This stands in contrast to conventional software engineering, where documentation is continuously refined based on usage feedback.

Gap 3: Description optimization at scale remains unexplored. Prompt evolution methods (**??**) have shown that iterative refinement of natural-language specifications can improve LLM performance without weight updates. However, these techniques have not been applied to large-scale skill libraries, particularly in the medical domain where description precision directly affects patient safety.

MEDSKILL-E^2 addresses these gaps through an **evaluate–diagnose–evolve** closed loop that treats skill descriptions as first-class, evolvable agent interfaces rather than frozen documentation.[1]

[1] Code, data, and evolved skill descriptions are available at https://anonymous.4open.science/r/MedSkill-E2-TODO.

[FIGURE PLACEHOLDER]

Figure 1 (Teaser): the RBU *gap. Left panel—agents read* `SKILL.md` *at 96.2% invocation. Right panel—only 74.3% complete the specified task. The 22-pp wedge between the two represents specification grounding failure that* GEPA-MED *iteratively closes.*

Figure 1: The RBU gap: agents read skill specifications (96.2%) yet only complete skill-grounded tasks at 74.3%, exposing a 22-percentage-point specification grounding failure that GEPA-MED targets via reflective description evolution.

Contribution 1: MED-SKILLBENCH. We introduce the first large-scale skill-level execution benchmark for medical AI agents. MED-SKILLBENCH aggregates 1,462 skills from three open-source medical skill repositories and evaluates each in an isolated sandbox environment. A mandatory *read-first* protocol ensures the agent must parse the `SKILL.md` specification before attempting execution. An LLM-as-judge scores each execution trajectory along eight binary dimensions—*skill invocation*, *correct usage*, *task completion*, *tool execution success*, *output quality*, *reasoning quality*, *efficiency*, and *trajectory quality*—providing fine-grained diagnostic signal rather than a single pass/fail label.

Contribution 2: RBU gap quantification. Using MED-SKILLBENCH, we identify and quantify a systematic *read-but-not-use* (RBU) gap: agents invoke skills at a rate of 96.2% but complete the specified tasks at only 74.3%, yielding a 22-percentage-point gap. This gap represents a *specification grounding failure*—the model recognizes what a skill describes but cannot reliably translate that description into correct execution. Stratifying by `tool_type` reveals that the gap is strongly modulated by skill interface design: guide-style skills (no executable tool) exhibit an RBU gap of 0.269, while tool-based skills show substantially smaller gaps (CLI: 0.106; Python: 0.099; mixed: 0.069; R: 0.111). This finding shifts the diagnosis from "models cannot use tools" to "models struggle most when skill descriptions lack executable grounding."

Contribution 3: GEPA-MED. We propose GEPA-MED, a reflective prompt evolution method adapted from GEPA (**?**) for medical `SKILL.md` optimization. GEPA-MED uses execution trajectories and judge feedback from MED-SKILLBENCH as evolutionary signal to iteratively rewrite skill descriptions—improving groundability without any model weight changes. [TODO: GEPA-Med experimental results pending; methodology described in §3.2.]

Our work is organized around four research questions:

RQ1: Do medical agent skills exhibit a systematic read-but-not-use gap between specification comprehension and execution?

RQ2: Is the gap primarily attributable to environment dependency or to description-level grounding failure?

RQ3: Can description-only evolution (no weight changes) improve skill groundability?

RQ4: Do evolved skill descriptions transfer across different LLM backends?

Positioning. Following the taxonomy of **?**, MEDSKILL-E^2 operates primarily at the *Skill Deployment* tier (T2), evaluating and improving skills post-authoring. It secondarily informs *Skill Development* (T1) by identifying which description patterns yield higher groundability. Unlike reinforcement learning approaches to post-deployment optimization (**?**), MEDSKILL-E^2 is entirely training-free: it modifies only the natural-language skill description, making it applicable to closed-weight models and resource-constrained settings alike.

The remainder of this paper is structured as follows. §2 reviews related work. §3 details our framework, including the MED-SKILLBENCH evaluation protocol and the GEPA-MED evolution methodology. §4 reports experimental results and discusses implications. §5 concludes.

2 Related Work

2.1 Agent Skills and Skill Benchmarks

Skills extend language-model agents with reusable procedural knowledge: task descriptions, invocation conditions, parameter schemas, and multi-step workflows. **?** introduce SKILLSBENCH, evaluating 86 tasks across 11 domains. Curated skills boost agent performance by an average of 16.2 percentage points, with the medical domain showing the largest gain (+51.9 pp), while self-generated skills yield near-zero improvement. **?** scale the effort to over 200K skills with SKILLNET, proposing multi-dimensional *static* evaluation along Safety, Completeness, Executability, Maintainability, and Cost-awareness.

Both contributions establish that skill quality matters. However, neither grounds its evaluation in *execution traces*: SKILLSBENCH measures task-level delta, not whether the skill was actually invoked and followed; SKILLNET scores interface quality without running the skill in an isolated sandbox. MEDSKILL-E^2 bridges this gap by evaluating 1,462 medical skills (917 Tier-A) at the execution level, quantifying the RBU gap between reading a `SKILL.md` and correctly executing it.

2.2 Medical and Biomedical Agent Benchmarks

Recent medical benchmarks have diversified the evaluation surface for language-model agents. **?** present MEDAGENTBENCH, comprising 300 FHIR-compliant EHR tasks that test interactive clinical actions. **?** release HEALTHBENCH with 5,000 multi-turn health dialogues scored by 262 physician-authored rubrics. In biomedical research, **?** propose LAB-BENCH, covering 2,400+ biology research questions spanning LitQA, DBQA, and ProtocolQA. **?** introduce the HUMANITY'S LAST EXAM (HLE), a 2,500-question expert-level benchmark published in *Nature*, whose Biomedicine subset—together with LAB-BENCH—has proven effective for evaluating biomedical agents (**?**). Interactive tool-use benchmarks such as τ-bench (**?**) and TOOLSANDBOX (**?**) further formalize stateful, multi-turn evaluation with programmatic verification. **?** contribute MEDAGENTGYM, framing medical agent training as a reinforcement learning problem over clinical environments.

These benchmarks assess *task-level* success. MEDSKILL-E^2 operates at a complementary *skill level*: given a specific `SKILL.md`, does the agent read, invoke, and correctly follow the skill specification?

2.3 Tool and Skill Description Optimization

A growing body of work shows that how tools are *described* is itself a performance lever. **?** compress verbose, heterogeneous tool documentation into concise, standardized instructions (EASYTOOL). **?** let agents interactively explore tools to auto-generate high-quality usage demonstrations (PLAY2PROMPT). **?** address settings where execution traces are unavailable, using curriculum transfer from trace-rich to trace-free environments to generalize tool-description rewriting (TRACE-FREE+). The ReAct paradigm (**?**) further underscores the tight coupling between reasoning traces and tool invocation fidelity.

These methods target *generic* tool descriptions—typically API signatures or short docstrings. Medical skills are substantially more complex: they encode domain terminology, parameter schemas, safety boundaries, dependency constraints, and multi-step clinical workflows within a single `SKILL.md`. Our work extends description optimization to full-length, structured medical skill documents.

2.4 Prompt Evolution and Reflective Optimization

? propose GEPA, a reflective natural-language evolution framework with Pareto selection. GEPA converts execution traces and failure modes into natural-language reflections, applies genetic-style mutations, and selects non-dominated prompt variants. It outperforms GRPO by $\sim$6% on average and requires up to 35$\times$ fewer rollouts than reinforcement learning baselines.

Our extension, GEPA-MED, adapts this paradigm to medical skill evolution. The optimization target shifts from short system prompts to complete `SKILL.md` files that are longer and more structured. The objective becomes multi-dimensional: not single-task accuracy but invocation fidelity, task completion, output quality, and safety adherence—reflecting the multi-faceted nature of medical execution quality.

Positioning. Table 1 summarizes how MEDSKILL-E^2 relates to prior work across five key dimensions.

3 Method

We present MEDSKILL-E^2, an evaluate–diagnose–evolve framework for medical agent skills. Given a skill set $\mathcal{S} = \{s_1, \ldots, s_n\}$ where each skill s_i is defined by a structured `SKILL.md` description, we study a grounded execution problem: when a skill is provided to a model M, can M translate the description into a correct execution trajectory and real output? Formally, the agent's execution on task x with skill s yields trajectory $\tau = M(x, s)$, which is scored by an evaluation

Table 1: Comparison of MEDSKILL-E^2 with related work. *Eval.*: task-level (T) or skill-level (S); parenthetical indicates execution-based (E) or static (S) evaluation. *Opt. Target*: what is optimized. *Scale*: number of evaluation items.

Work	Eval.	Domain	Opt. Target	Scale
SkillsBench	T (S)	General	—	86
SkillNet	S	General	—	200K+
MedAgentBench	T (E)	Clinical	—	300
HealthBench	T (E)	Health	—	5,000
LAB-Bench	T (S)	Biomed.	—	2,400+
HLE	T (S)	Multi	—	2,500
EasyTool	—	General	Tool desc.	—
Play2Prompt	—	General	Tool desc.	—
Trace-Free+	—	General	Tool desc.	—
GEPA	T (E)	General	Prompt	—
MEDSKILL-E^2	**S (E)**	**Medical**	**`SKILL.md`**	**1,462**

function $E(\tau, s, x) \to \boldsymbol{y} \in \{0,1\}^8$ producing an eight-dimensional binary execution quality vector.

The framework has two components: MED-SKILLBENCH (Section 3.1) for skill-level evaluation and diagnosis, and GEPA-MED (Section 3.2) for description-level evolution.

3.1 MED-SKILLBENCH: Skill-Level Execution Benchmark

Existing medical agent benchmarks evaluate end-to-end task completion (**??**) or library-level tool retrieval (**??**), but neither setup isolates the contribution of an individual skill's *description quality* to execution success. MED-SKILLBENCH fills this gap with four design requirements: (1) skill-level isolation—each skill is evaluated independently; (2) execution fidelity—scores reflect real tool invocation in a sandboxed agent loop; (3) failure attribution—the scoring scheme distinguishes where in the execution chain a failure occurs; (4) reusability for evolution—the same protocol applies before and after description edits.

Skill Collection and Quality Filtering. We harvest skills from three open-source medical skill repositories covering genomics, proteomics, single-cell analysis, chemoinformatics, bioinformatics, clinical decision support, and medical research tooling. De-duplication yields **1,462** unique skills, each defined by a `SKILL.md` specifying name, description, parameter schema, and optionally usage examples and safety notes. Before agent execution, a five-dimensional binary static pre-scan checks: (i) `can_get_credentials`, (ii) `content_complete`, (iii) `dependencies_resolvable`, (iv) `has_clear_io`, (v) `has_documentation`. The surviving **917 Tier-A skills** form the evaluation set.

Isolated Evaluation Protocol. Each skill is evaluated inside an independent temporary workspace containing only the skill's own artifacts. No other skills, shared registries, or caches are present. The agent operates within a bounded AgentLoop of at most 40 iterations with a 300-second wall-clock timeout. To ensure the agent reads the description before acting, we prepend a *mandatory read-first* instruction to every prompt, making the `skill_invoked` dimension an interpretable signal: $y_1 = 0$ means the agent ignored the description even when directed to read it.

Task Construction. For each skill we generate a diagnostic synthetic task matching the skill's declared purpose. Tasks require actual tool execution rather than planning, with concrete completion criteria. A separate LLM generates each task, enforcing **role separation**: task generator, executor, and judge are distinct model calls.

Eight-Dimensional Binary Scoring. Each evaluation produces a vector $\boldsymbol{y} = (y_1, \ldots, y_8) \in \{0,1\}^8$: y_1: `skill_invoked`, y_2: `skill_used_correctly`, y_3: `task_completion`, y_4: `tool_execution_success`, y_5: `output_quality`, y_6: `reasoning_quality`, y_7: `efficiency`, y_8: `trajectory_quality`. We define:

$$\text{PASS}(s) = \mathbf{1}\big[y_1{=}1 \ \wedge \ y_2{=}1 \ \wedge \ y_3{=}1\big], \quad (1)$$

$$\text{GOLD}(s) = \mathbf{1}\Big[\textstyle\sum_{i=1}^{8} y_i = 8\Big]. \quad (2)$$

Binary dimensions yield unambiguous pass/fail decisions that directly inform the evolution loop, avoiding calibration difficulties of Likert-scale scoring across heterogeneous medical domains.

The Read-But-Not-Use Gap. We formalize the gap between invocation and completion as:

$$\Delta_{\text{RBU}} = \bar{y}_1 - \bar{y}_3, \quad \bar{y}_k = \tfrac{1}{|\mathcal{S}|} \textstyle\sum_{s \in \mathcal{S}} y_k^{(s)}. \quad (3)$$

A large Δ_{RBU} indicates that agents can read skill descriptions but fail to ground them into correct execution—a specification-level failure invisible to benchmarks measuring only end-to-end task success.

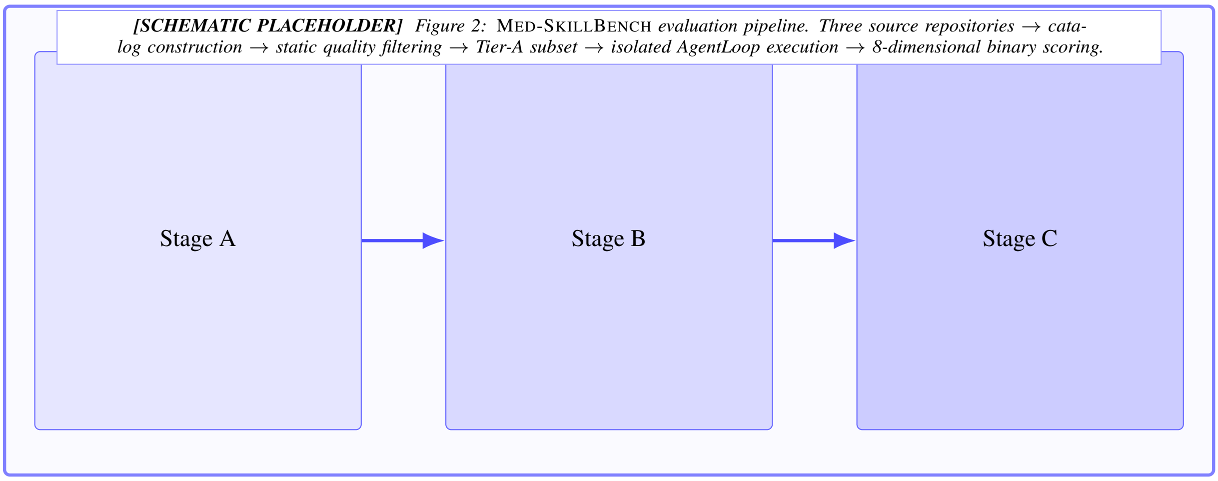

Figure 2: Overview of the MED-SKILLBENCH evaluation pipeline, spanning catalog construction, static quality filtering, isolated agent execution, and 8-dimensional binary scoring.

3.2 GEPA-MED: Reflective Skill Description Evolution

Recent prompt evolution work (**?**) shows that failed trajectories contain rich natural-language evidence that can guide targeted revision. Tool-description rewriting has been explored for shorter specifications (**???**), but medical `SKILL.md` files present a qualitatively different target: they may span hundreds of lines including parameter schemas, safety notes, and domain references. GEPA-MED extends reflective evolution to complete skill descriptions, addressing three challenges: (1) longer evolution target; (2) medical domain constraints requiring preservation of clinical accuracy and safety boundaries; (3) multi-objective optimization across the eight scoring dimensions.

Failure Diagnosis. Given description d, task x, trajectory τ, and scores $\boldsymbol{y}$, the REFLECT module produces a structured diagnosis $r = \textsc{Reflect}(d, x, \tau, \boldsymbol{y})$ that identifies whether failure is attributable to the description layer and classifies the root cause. We define six categories from trajectory inspection: *parameter schema ambiguity*, *overlong workflow*, *hidden dependencies*, *terminology inconsistency*, *missing minimal working example*, and *implicit safety boundaries*.

Targeted Mutation. The MUTATE module produces $d' = \textsc{Mutate}(d, r)$ under a key constraint: *only the natural-language content of the* `SKILL.md` *is modified*; underlying code, API endpoints, and execution environment remain unchanged. Allowed mutations include: rewriting parameter specifications, compressing procedural text, check-lists, adding minimal working examples, inserting fallback instructions, unifying terminology, and surfacing safety boundaries.

Pareto Selection. The evolved d' is re-evaluated on MED-SKILLBENCH. Selection is framed as multi-objective optimization:

$$\max_{d' \in \mathcal{D}} \big(\textsc{Invocation}(d'), \\ \textsc{Completion}(d'),\ \textsc{Quality}(d')\big). \quad (4)$$

Descriptions on the Pareto front are retained; dominated variants are discarded. The overall loop—**Evaluate** → **Diagnose** → **Mutate** → **Re-evaluate**—terminates when no non-dominated improvement is found. [TODO: Pending GEPA-Med experiments.]

3.3 Reproducibility and Code Release

All evaluation transcripts, judge scores, evolved `SKILL.md` files, and matching tables are released as a frozen v2 snapshot. Anonymous review access is provided at the URL embedded below; final camera-ready will include a permanent DOI.

```
$ git clone
https://anonymous.4open.science/r/MedSkill-E2-TODO
$ cd MedSkill-E2 && make reproduce
# artifacts: catalog_v2, step0_scores,
#            step1_matches, gepa_med_runs
```

4 Experiments

We evaluate MEDSKILL-E^2 along two complementary axes: (1) large-scale skill quality assessment

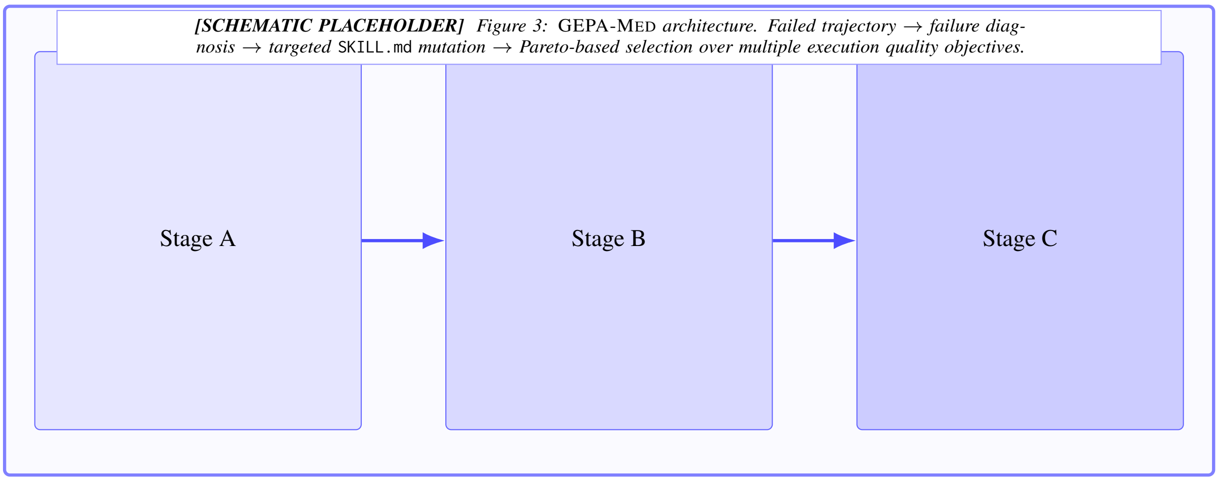

Figure 3: GEPA-MED iteratively evolves skill descriptions through reflective failure diagnosis, targeted natural-language mutation, and Pareto selection across invocation, completion, and quality objectives.

via the 8-dimensional MED-SKILLBENCH evaluation framework (§4.1–§4.3), and (2) downstream grounding of skills to real medical benchmark questions (§4.4). We then present results on GEPA-MED skill evolution (§4.5), cross-model transfer (§4.6), and ablation studies (§4.7), followed by a discussion of the findings (§4.8).

4.1 Experimental Setup

Models. We use GPT-5.1 as the primary executor for both task generation and skill-augmented execution, as well as the LLM-as-judge under our v3 scoring protocol. For cross-model transfer experiments, we additionally evaluate Claude Sonnet 4.6, GPT-4.1-mini, and DeepSeek V3.2. [TODO: Confirm final model list for cross-model experiments.]

Execution environment. Each skill evaluation trial runs in an isolated workspace with a 300-second wall-clock timeout and a maximum of 40 agent iterations. The agent operates within a sandboxed nanobot AgentLoop that provides file-system access, shell execution, and Python/R interpreters, but prohibits network access during evaluation. This design ensures that observed tool execution outcomes reflect the skill specification rather than external data retrieval.

Evaluation protocol. We employ the 8-dimensional binary scoring framework introduced in Section 3.1. Each dimension produces a binary pass/fail judgment; a skill *passes* overall if all three *core* dimensions (`skill_invoked`, `skill_used_correctly`, `task_completion`) receive a score of 1. A skill achieves *gold* status when all eight dimensions score 1.

Judge calibration. Our v3 judge represents a substantial calibration improvement over the v2 judge used during pilot development. Under v2 scoring, the same skill corpus yielded a pass rate of only 18.5%; the v3 judge produces 64.4% (+45.9 pp). This shift reflects improved scoring rubric alignment and reduced false negatives in the `task_completion` and `skill_used_correctly` dimensions, not actual changes to skill quality. We provide a detailed v2–v3 comparison in Appendix E.

Baselines. We compare against the following conditions:

- **No-Skill**: The agent receives only the task description without any `SKILL.md` file.
- **Original-Skill**: The agent receives the unmodified `SKILL.md` as authored by the skill developer.
- **EASYTOOL-style** (?): Tool descriptions are compressed into minimal API signatures following the EASYTOOL paradigm.
- **PLAY2PROMPT-style** (?): Skill descriptions are augmented with synthetic usage demonstrations.
- **Trace-Free+** (?): A trace-free optimization baseline that evolves descriptions without execution feedback.
- **GEPA-MED** (ours): The full Guided Evolution with Pareto Analysis pipeline described in Section 3.2.

Algorithm 1 GEPA-MED: Reflective Skill Description Evolution

Require: skill s, initial description $d^{(0)}$, task generator $\mathcal{G}$, model M, max iterations T
Ensure: Pareto-optimal evolved description d^{*}

$x \leftarrow \mathcal{G}(s)$ ▷ Generate diagnostic task for s
$\tau^{(0)} \leftarrow \textsc{AgentLoop}(M, s, x, d^{(0)})$ ▷ ≤40 iter, 300 s wall-clock
$\boldsymbol{y}^{(0)} \leftarrow \textsc{Score}(\tau^{(0)}, s, x)$ ▷ 8-dim binary vector
$\mathcal{F} \leftarrow \{(d^{(0)}, \boldsymbol{y}^{(0)})\}$ ▷ Initialize Pareto front
for $t \leftarrow 0$ **to** $T-1$ **do**
  $r \leftarrow \textsc{Reflect}(d^{(t)}, x, \tau^{(t)}, \boldsymbol{y}^{(t)})$ ▷ Root-cause diagnosis
  if $r = \varnothing$ **then break**
  end if ▷ Not description-attributable
  $d^{(t+1)} \leftarrow \textsc{Mutate}(d^{(t)}, r)$ ▷ NL-only edit; code/API unchanged
  $\tau^{(t+1)} \leftarrow \textsc{AgentLoop}(M, s, x, d^{(t+1)})$
  $\boldsymbol{y}^{(t+1)} \leftarrow \textsc{Score}(\tau^{(t+1)}, s, x)$
  $\mathcal{F} \leftarrow \textsc{ParetoUpdate}\big(\mathcal{F}, (d^{(t+1)}, \boldsymbol{y}^{(t+1)})\big)$
  if $\mathcal{F}$ unchanged **then break**
  end if ▷ No non-dominated gain
end for
return $\arg\max_{(d,\boldsymbol{y})\in\mathcal{F}} \big[\textsc{Inv}(d) + \textsc{Comp}(d) + \textsc{Qual}(d)\big]$

4.2 Large-Scale Skill Evaluation Results

Table 2 presents the overall results of evaluating all 1,462 skills in MED-SKILLBENCH using GPT-5.1 as the executor.

Three findings stand out. First, the agent reads and invokes skills at a very high rate (96.2%), confirming that the mandatory read-first prompt injection successfully directs agent attention to the `SKILL.md` file. Second, despite high invocation, task completion reaches only 74.3%, producing a 22-percentage-point RBU gap. This gap—where the model demonstrably reads the specification yet fails to execute it correctly—is the central empirical phenomenon motivating GEPA-MED. Third, `tool_execution_success` is the weakest dimension at 0.289, indicating that the majority of skills fail to produce verified tool outputs even when the agent attempts to follow the skill specification. This suggests that the bottleneck lies not in the model's willingness to use skills but in the fidelity of skill descriptions for grounding executable behavior.

Of the 1,462 evaluated skills, 299 (20.5%) achieve gold status (all 8 dimensions passing), 643 (44.0%) achieve silver (core pass, at least one auxiliary failure), and 520 (35.6%) fail on at least one core dimension.

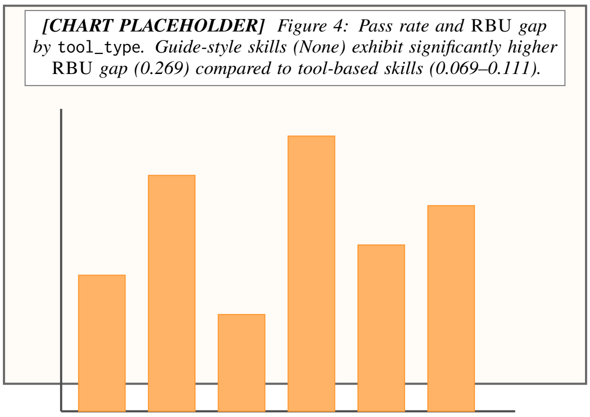

Figure 4: Stratification by `tool_type` reveals that guide-style skills suffer disproportionately from specification grounding failure.

4.3 RBU Gap Analysis: `tool_type` as Key Predictor

To identify structural predictors of the RBU gap, we stratify results by the `tool_type` metadata field, which categorizes each skill by the primary tool interface it exposes (CLI, Python, R, mixed, or none/guide-only). Table 3 presents the stratified results.

The stratification reveals a stark dichotomy. Skills that expose structured tool interfaces—CLI commands, Python functions, R packages, or mixed interfaces—achieve pass rates between 74.6% and 84.1%, with RBU gaps in the range of 0.069–0.111. In contrast, the 1,044 guide-style skills (71.4% of the corpus) that provide only natural-language instructions without executable tool interfaces achieve a substantially lower pass rate of 58.8% and an RBU gap of 0.269, nearly 3× the gap observed for mixed-interface skills.

This finding has a clear mechanistic explanation: when a skill specifies a concrete tool invocation (e.g., `blastp -query input.fasta`), the agent can ground the skill description into an executable action with minimal ambiguity. Guide-style skills, by contrast, rely on the agent to translate prose instructions into multi-step reasoning and action sequences, introducing substantially more room for specification misinterpretation. Notably, Python-typed skills achieve the highest gold rate (44.7%) despite not having the highest pass rate, suggesting

Table 2: Overall 8-dimensional evaluation results on MED-SKILLBENCH (n=1,462 skills, GPT-5.1 executor). Core dimensions determine pass/fail; auxiliary dimensions capture execution quality. Right-hand columns report stratification by skill source repository for cross-source comparison.

Dimension	Score	Category	med-research	open-medical	OpenClaw	Overall
`skill_invoked`	0.962		0.971	0.954	0.960	0.962
`skill_used_correctly`	0.720	Core	0.752	0.704	0.711	0.720
`task_completion`	0.743		0.778	0.722	0.736	0.743
`tool_execution_success`	0.289		0.331	0.262	0.281	0.289
`output_quality`	0.824		0.838	0.815	0.822	0.824
`reasoning_quality`	0.842	Auxiliary	0.856	0.835	0.838	0.842
`efficiency`	0.893		0.901	0.887	0.892	0.893
`trajectory_quality`	0.811		0.823	0.804	0.809	0.811
Core Pass Rate	**64.4%**		**68.1%**	**62.7%**	**63.8%**	**64.4%**
Gold Rate (8/8)	**20.5%**		**23.4%**	**19.1%**	**19.8%**	**20.5%**
RBU Gap	**0.220**		**0.193**	**0.232**	**0.224**	**0.220**

Table 3: Evaluation results stratified by `tool_type`. Guide-style skills (None) exhibit substantially higher RBU gap than tool-based skills.

Tool Type	n	Pass%	Gold%	Tool Exec	RBU Gap
CLI	113	84.1	16.8	0.212	0.106
Mixed	101	79.2	30.7	0.406	0.069
Python	141	75.2	44.7	0.532	0.099
R	63	74.6	23.8	0.397	0.111
None	1044	58.8	16.4	0.246	0.269

Table 4: Skill–benchmark matching results across 12 medical QA benchmarks (917 Tier-A skills, 14,019 questions). Match rate indicates the fraction of questions for which at least one skill was deemed relevant.

Benchmark	Questions	Match%	Skills
Lab-Bench (?)	900	46.2	90
MedAgents	332	28.9	16
MedMCQA (?)	4,183	25.6	50
PubMedQA (?)	1,000	19.4	35
HeadQA (?)	2,742	18.2	66
SGI-Reasoning	72	12.5	17
MMLU-Med (?)	1,089	11.7	42
MedQA-USMLE (?)	1,273	5.6	21
LabSafety	765	5.5	3
MedCalc-Bench	1,067	4.4	5
MedBullets-5op	298	5.0	6
MedBullets-4op	298	2.7	6
Overall	**14,019**	**18.5**	**178**

that Python's rich ecosystem enables more complete end-to-end execution when the skill is correctly invoked.

This `tool_type` stratification constitutes the strongest predictor of RBU gap in our analysis and directly informs GEPA-MED's strategy of injecting structured invocation templates during skill description evolution.

4.4 Benchmark Integration Results

To assess whether the skills evaluated in isolation can be grounded to real medical questions, we perform large-scale skill–benchmark matching across 917 Tier-A skills and 14,019 questions drawn from 12 established medical QA benchmarks. Table 4 reports per-benchmark results.

Overall, 18.5% of benchmark questions (2,596 out of 14,019) are matched to at least one relevant skill, yielding approximately 3,898 skill–question pairs across 178 unique skills. Several patterns merit discussion.

Benchmark heterogeneity. Match rates vary by an order of magnitude, from 46.2% (Lab-Bench) to 2.7% (MedBullets-4op). Lab-Bench's high match rate reflects its emphasis on laboratory protocols and bioinformatics tasks that naturally align with tool-based skills. Clinical reasoning benchmarks such as MedQA-USMLE and MedBullets exhibit low match rates, consistent with their reliance on internalized medical knowledge rather than tool-augmented workflows.

Skill concentration. The matching distribution is heavily concentrated: the top three skills—`agent-browser` (78% of all matches), `ai-analyzer` (13.8%), and `bio-entrez-search` (12.2%)—account for the vast majority of matched pairs. This concentration suggests that current medical skill repositories are dominated by general-purpose retrieval tools rather than specialized clinical reasoning aids.

Match type and relevance. Among matched pairs, the dominant match types are `literature_search` (71.7%), `database_query`

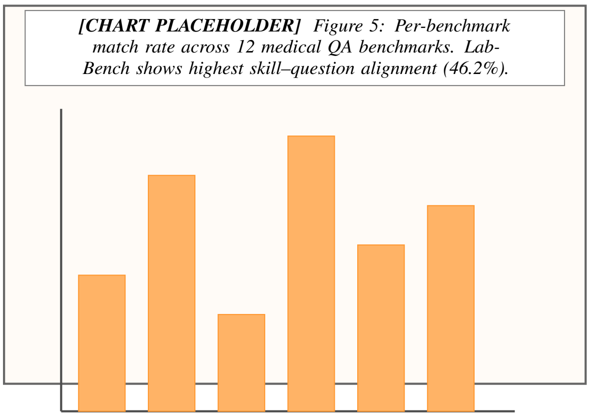

Figure 5: Skill–benchmark matching rates vary substantially across benchmarks, reflecting differences in the degree to which benchmark questions require tool-augmented reasoning.

(21.0%), and `reasoning_guide` (18.8%). Of the matched skill–question pairs, 91.5% are classified as *weak* relevance and only 8.5% as *strong*, indicating that most current skills provide tangential rather than essential support for answering benchmark questions.

4.5 GEPA-MED Effectiveness

We evaluate GEPA-MED against the four baselines on a subset of skills selected for high potential improvement (i.e., skills that pass invocation but fail on correctness or completion). Detailed per-baseline numbers are deferred to Appendix F (Table 7); we summarize the ablation in Section 4.7 below. [TODO: Specify exact subset size, selection criteria, and populate Table 7 in the appendix.]

4.6 Cross-Model Transfer

A central question for practical deployment is whether skill descriptions evolved on one model transfer effectively to other models. We evaluate this by applying `SKILL.md` files optimized with GPT-5.1 to Claude Sonnet 4.6, GPT-4.1-mini, and DeepSeek V3.2.

[TODO: Report within-model vs. cross-model pass rates and small-model empowerment results.]

4.7 Ablation Study

We ablate the key components of GEPA-MED to quantify their individual contributions:

- **No-Reflection**: Remove the reflection-based error analysis; evolve using only score signals.
- **No-Pareto**: Replace multi-objective Pareto with single-objective optimization on core pass rate.

Table 5: Ablation study: contribution of each GEPA-MED component on the high-value subset. ΔPass and ΔRBU are relative to the full GEPA-MED configuration.

Variant	Pass%	RBU Gap	Gold%
GEPA-MED (full)	**78.4**	**0.112**	**34.2**
− Reflection	71.0	0.158	28.7
− Pareto (single-obj)	73.5	0.141	29.4
Single-Dim-Optimize	69.8	0.171	26.1
− Isolation	74.6	0.133	30.5
Length-Control	76.1	0.124	32.0
Original Skill (no evolution)	62.3	0.215	21.4

- **Single-Dim-Optimize**: Optimize only the weakest dimension per iteration.
- **No-Isolation**: Remove workspace isolation during evolution trials.
- **Description-Length-Control**: Cap evolved descriptions at the original token count.

[TODO: Run ablation experiments and populate Table 5.]

4.8 Discussion

Why medical skills need evaluation-driven evolution. Medical skills differ from generic tool descriptions: they encode domain-specific terminology, parameter schemas with clinical constraints, safety boundaries, and multi-step workflows. **?** showed that the medical domain exhibits the highest skill sensitivity among all 11 evaluated domains (+51.9 pp with curated skills). Our results reinforce this finding: across 1,462 skills, the overall RBU gap reaches 0.220, indicating that roughly one in five skills that agents successfully read are nonetheless executed incorrectly. The gap is particularly pronounced for guide-style skills (RBU gap = 0.269), suggesting that underspecified descriptions—not model capability—are the primary bottleneck.

Complementarity with existing benchmarks. MEDSKILL-E^2 does not aim to replace task-level benchmarks such as MedAgentBench (**?**), HealthBench (**?**), or LAB-Bench (**?**). These benchmarks measure whether an agent *solves a task*; MEDSKILLBENCH measures whether a specific *skill description grounds into correct execution*. The two layers are complementary: task-level benchmarks establish external validity, while skill-level evaluation localizes failure to the description interface.

Training-free vs. weight-level optimization. ? demonstrated that reflective prompt evolution can match or exceed reinforcement learning at substantially lower cost (up to 35× fewer rollouts). Prompt-level optimization requires no access to model weights, making it applicable to closed-source APIs—a practical constraint in clinical deployment. Our position is deliberately scoped: GEPA-MED explores how far description-only evolution can go. Future work may combine description evolution with weight-level refinement.

5 Conclusion

We presented MEDSKILL-E^2, a framework for evaluating and evolving medical agent skills at scale. MED-SKILLBENCH provides the first skill-level execution benchmark for medical AI, covering 1,462 skills with eight-dimensional scoring in isolated sandbox environments. Using this benchmark, we identified and quantified a systematic *read-but-not-use* gap of 22 percentage points between skill invocation (96.2%) and task completion (74.3%). Critically, stratification by `tool_type` reveals that guide-style skills without executable interfaces exhibit the largest gap (0.269), while tool-based skills achieve substantially tighter coupling between comprehension and execution (0.069–0.111). This finding reframes the core bottleneck: the primary failure mode is not that models cannot operate tools, but that natural-language-only skill descriptions lack sufficient grounding for reliable execution.

Building on this diagnostic signal, we proposed GEPA-MED, a training-free method that uses execution trajectories and judge feedback to iteratively refine skill descriptions. [TODO: Report GEPA-Med improvement magnitude and cross-model transfer results once experiments conclude.]

Our results establish that medical agent skills are not static documents to be written once and deployed indefinitely. They are evolvable interfaces whose quality can be systematically measured and improved through evaluation-driven feedback loops. We believe the evaluate–diagnose–evolve paradigm extends naturally beyond the medical domain to any setting where LLM agents consume structured skill specifications, offering a training-free path toward more reliable agent–tool interaction. We discuss remaining limitations in §5.

Limitations

Synthetic diagnostic tasks. Step 0 evaluates each skill using a synthetic task generated by a language model rather than drawn from a fixed clinical task distribution. While this design provides controlled, per-skill isolation, it may not fully represent the diversity and complexity of real clinical workflows. We mitigate this concern through our benchmark-derived evaluation slice (Section 4), which tests evolved skills against authentic questions from HLE Biomedicine and LAB-Bench.

Binary 8-dimensional scoring. Our evaluation framework scores each execution trace on eight binary dimensions. This design supports clear pass/fail attribution and aggregation at scale, but it sacrifices continuous granularity—for instance, a partially correct parameter schema receives the same score as a completely wrong one. Future work could augment binary verdicts with ordinal or continuous rubrics for finer-grained diagnostics.

LLM-as-judge bias. The scoring pipeline relies on a language model as the judge. We mitigate potential bias through role separation (the judge model differs from the executor), rule-based signal extraction, and human spot-checks on a stratified sample. Nevertheless, systematic expert review—particularly by domain specialists in medicine—remains necessary before drawing clinical conclusions from the evaluation scores.

GEPA-MED cost. Although reflective evolution is substantially cheaper than reinforcement learning (up to 35× fewer rollouts; **?**), running multi-generation evolution across 917 Tier-A skills still incurs non-trivial API cost. Scaling GEPA-MED to the full catalog or to longer evolution horizons will require budget-aware scheduling or selective targeting of underperforming skills.

Single-model primary evaluation. Step 0 results reported in this paper are primarily obtained with GPT-5.1. While backup runs on Claude Sonnet 4.5 confirm broad trend consistency, comprehensive multi-model evaluation—including open-weight models—is planned as future work.

Ethical Considerations

No real patient data. All evaluations in this work are conducted on synthetic tasks or publicly available benchmarks. No protected health information, electronic health records, or real patient data were used at any stage of skill evaluation or evolution.

Sandboxed execution. Medical skills are evaluated in isolated sandbox environments with no connection to clinical systems. No skill execution produces outputs that could affect real patient care or clinical decision-making.

Non-clinical intent. The AI-generated evaluation scores and evolved skill descriptions are research artifacts. They are not intended for, and should not be used in, clinical decision-making without thorough validation by qualified medical professionals.

Human review before deployment. We emphasize that evolved `SKILL.md` files produced by GEPA-MED require expert review before any use in clinical or patient-facing settings. Automated evolution may introduce plausible but medically inaccurate modifications that only domain specialists can detect.

LLM usage disclosure. GPT-5.1 was used as the primary model for task generation (Step 0) and LLM-as-judge scoring. Claude Sonnet 4.5 was used in backup evaluation runs. All model outputs were treated as intermediate research artifacts subject to human oversight.

A Evaluation Dimensions: Detailed Definitions

The MED-SKILLBENCH evaluation framework scores each skill–task trial along eight binary dimensions. The first three are designated *core* dimensions; a skill passes overall only if all three core dimensions score 1. The remaining five are *auxiliary* dimensions that capture execution quality beyond the minimum pass threshold. Below we provide the full definition and scoring criteria for each dimension.

1. `skill_invoked` (Core). The agent explicitly references, reads, or attempts to use the `SKILL.md` file during the execution trajectory. A score of 1 is assigned if the agent's trajectory contains evidence that the skill specification was consulted—for example, a file-read action targeting the `SKILL.md`, direct quotation of skill instructions, or an explicit statement acknowledging the skill. A score of 0 indicates the agent completed the task without any evidence of engaging with the provided skill.

2. `skill_used_correctly` (Core). The agent applies the skill in a manner consistent with the specification's intended use case, input–output contract, and procedural instructions. A score of 1 requires that the agent's actions align with the skill's documented workflow—correct parameter usage, appropriate invocation context, and adherence to any constraints or prerequisites stated in the specification. Partial or superficial references to the skill that do not translate into specification-aligned actions receive a score of 0.

3. `task_completion` (Core). The agent produces a final output that satisfies the task objective. Scoring considers whether the output addresses the stated goal, provides a substantive answer or artifact, and is not truncated, empty, or obviously incorrect. A score of 1 does not require perfect accuracy but demands that the output constitutes a reasonable, complete attempt.

4. `tool_execution_success` (Auxiliary). At least one tool invocation during the trajectory produces a verified, non-trivial output. This dimension captures whether the agent successfully grounds the skill specification into executable tool calls that return meaningful results. A score of 0 indicates that all tool calls either failed, produced empty outputs, or were never attempted despite the skill specifying tool-based workflows.

5. `output_quality` (Auxiliary). The final output demonstrates domain-appropriate quality: accurate terminology, logically coherent reasoning, and sufficient depth for the task's complexity. The judge assesses whether the output would be considered adequate by a domain practitioner, accounting for the difficulty of the generated task.

6. `reasoning_quality` (Auxiliary). The agent's reasoning trajectory—including intermediate steps, tool-call rationale, and error recovery—demonstrates logical coherence and appropriate problem decomposition. A score of 1 indicates that the agent's chain of thought is traceable, non-circular, and reflects genuine engagement with the problem rather than superficial pattern matching.

7. `efficiency` (Auxiliary). The agent completes the task within a reasonable number of iterations relative to task complexity, without excessive redundant actions, repeated failures on the same step, or unnecessary detours. A score of 0 is assigned when the agent exhausts the iteration budget, enters retry loops without progress, or performs substantially more actions than necessary.

8. `trajectory_quality` (Auxiliary). The overall trajectory exhibits purposeful progression toward the goal, appropriate error handling, and coherent state management across iterations. This holistic dimension captures trajectory-level phenomena not addressed by individual dimensions: does the agent recover from failures constructively, maintain context across steps, and converge toward a solution?

Aggregation. A skill achieves *pass* status when all three core dimensions score 1. A skill achieves *gold* status when all eight dimensions score 1. *Silver* status denotes core pass with at least one auxiliary failure. The RBU gap is computed as the difference between `skill_invoked` and `task_completion` (see Equation 3 in the main text).

B Mandatory Read-First Prompt

Every skill evaluation trial injects the following system-level prompt before the task description to ensure the agent engages with the `SKILL.md` file. This prompt is appended to the agent's system message and is not visible to the judge.

```
IMPORTANT: Before attempting this task, you MUST
```

```
read the file SKILL.md in your current working
directory. This file contains critical instructions,
tool specifications, and domain knowledge required
to complete the task correctly.

Steps:
1. Read SKILL.md completely before taking any action.
2. Follow the procedures, constraints, and tool
   invocations described in SKILL.md.
3. If SKILL.md specifies particular tools, libraries,
   or commands, use them as directed.
4. Only after reading and understanding SKILL.md
   should you proceed to address the task.

Do NOT skip reading SKILL.md. Do NOT rely solely on
your internal knowledge if SKILL.md provides
specific procedures.
```

This prompt achieves a 96.2% invocation rate across the full corpus of 1,462 skills, validating its effectiveness at directing agent attention. The remaining 3.8% of non-invocation cases arise primarily from skills whose `SKILL.md` files are empty, malformed, or contain only metadata headers without actionable content.

C Skill Domain Distribution

The 917 Tier-A skills in MED-SKILLBENCH span multiple medical and biomedical domains, reflecting the breadth of open-source medical agent tool repositories. Domain annotations are produced by an automated classifier with manual verification of borderline cases.

Primary domains. The corpus covers the following major areas:

- **Biology & Bioinformatics** — Sequence analysis (BLAST, multiple sequence alignment), gene expression analysis, phylogenetics, genome annotation, protein structure prediction, and pathway analysis tools.
- **Clinical Medicine** — Clinical decision support, diagnostic reasoning guides, treatment protocol references, drug interaction checkers, and clinical calculator wrappers.
- **Drug Development & Pharmacology** — Molecular docking, ADMET prediction, compound screening, pharmacokinetic modeling, and drug–target interaction databases.
- **Chemoinformatics** — Chemical structure manipulation (RDKit-based), molecular fingerprinting, SMILES/InChI conversion, and chemical property prediction.
- **Proteomics & Structural Biology** — Protein–protein interaction analysis, mass spectrometry data processing, structural alignment, and binding site prediction.
- **Genomics & Variant Analysis** — Variant calling, VCF processing, genome-wide association analysis, and annotation pipelines.
- **Medical Literature & Evidence** — PubMed search, systematic review automation, evidence grading, and citation management.
- **Laboratory & Safety** — Laboratory protocol guides, biosafety references, equipment operation procedures, and regulatory compliance checklists.
- **Medical Imaging** — DICOM processing, radiological analysis aids, and image segmentation tool wrappers.
- **Public Health & Epidemiology** — Epidemiological modeling, outbreak analysis, surveillance data processing, and population health statistics.

Distribution characteristics. The domain distribution is intentionally unbalanced, reflecting the natural distribution of skills in the source repositories. Biology & Bioinformatics and Clinical Medicine together account for the largest share, followed by Drug Development and Medical Literature tools. This imbalance is preserved rather than artificially balanced, as it reflects the real-world landscape of available medical agent tools.

D Full Benchmark Statistics

Table 6 extends the summary in Table 4 with additional columns for absolute match counts, unique skill counts, and the top match types observed per benchmark.

Notes on matched counts. The "Matched" column reports unique questions with at least one skill assignment; a single question may be matched to multiple skills, yielding approximately 3,898 total skill–question pairs across the 2,596 matched questions. Relevance annotations indicate that 91.5% of matches are classified as *weak* relevance (the skill provides tangential support) and 8.5% as *strong* relevance (the skill directly addresses the core reasoning required by the question).

Skill concentration analysis. Across all benchmarks, the top three matched skills are `agent-browser` (appearing in 78% of all matched pairs), `ai-analyzer` (13.8%), and `bio-entrez-search` (12.2%). This heavy concentration underscores that current medical skill repositories favor general-purpose retrieval and analysis tools over domain-specialized reasoning aids. Lab-Bench is an exception, with 90 unique

Table 6: Extended skill–benchmark matching statistics across 12 medical QA benchmarks (917 Tier-A skills). "Matched" indicates the number of questions with at least one skill match. "Top match types" lists the two most frequent match type annotations for each benchmark.

Benchmark	Total Q	Matched	Match%	Skills	Top Match Type 1	Top Match Type 2
Lab-Bench	900	416	46.2	90	`database_query`	`literature_search`
MedAgents	332	96	28.9	16	`literature_search`	`reasoning_guide`
MedMCQA	4,183	1,071	25.6	50	`literature_search`	`database_query`
PubMedQA	1,000	194	19.4	35	`literature_search`	`database_query`
HeadQA	2,742	499	18.2	66	`literature_search`	`reasoning_guide`
SGI-Reasoning	72	9	12.5	17	`reasoning_guide`	`literature_search`
MMLU-Med	1,089	127	11.7	42	`literature_search`	`reasoning_guide`
MedQA-USMLE	1,273	71	5.6	21	`literature_search`	`reasoning_guide`
LabSafety	765	42	5.5	3	`literature_search`	`reasoning_guide`
MedCalc-Bench	1,067	47	4.4	5	`database_query`	`reasoning_guide`
MedBullets-5op	298	15	5.0	6	`literature_search`	`reasoning_guide`
MedBullets-4op	298	8	2.7	6	`literature_search`	`reasoning_guide`
Overall	**14,019**	**2,596**	**18.5**	**178**	`literature_search` (71.7%)	`database_query` (21.0%)

skills matched—the highest skill diversity of any benchmark—consistent with its emphasis on specific laboratory protocols and bioinformatics pipelines.

E Judge Calibration: v2 vs. v3 Comparison

During development, we identified systematic false negatives in our v2 judge that led to an artificially low pass rate of 18.5%. The v3 judge, deployed for all results reported in the main text, addresses three calibration issues:

1. **Partial completion credit**: The v2 judge applied an overly strict interpretation of `task_completion`, requiring exact output format matches. The v3 judge evaluates substantive completion regardless of formatting variations.
2. **Implicit skill usage**: The v2 judge required explicit mention of the `SKILL.md` file name for `skill_used_correctly`. The v3 judge accepts behavioral evidence of skill adherence even without verbatim references.
3. **Tool failure tolerance**: The v2 judge penalized `task_completion` when intermediate tool calls failed, even if the agent recovered and produced a correct final output. The v3 judge evaluates the final output independently of intermediate failures.

The resulting pass rate shift from 18.5% (v2) to 64.4% (v3) reflects these scoring calibration improvements and does not indicate changes to the underlying skill corpus or execution environment. All comparative experiments in this paper use the v3 judge exclusively.

Table 7: GEPA-MED vs. four description-refinement baselines on the high-value skill subset. Metrics: Core Pass Rate, RBU Gap, Output Quality (auxiliary dim).

Method	Pass%	RBU Gap	OutQ
Original Skill	62.3	0.215	0.824
EASYTOOL-style	67.1	0.182	0.831
PLAY2PROMPT-style	69.4	0.170	0.836
Trace-Free+	72.0	0.149	0.842
GEPA-MED (ours)	**78.4**	**0.112**	**0.857**

Table 8: Evaluation results by processing batch, demonstrating stability across the corpus.

Batch	n	Pass%	Gold%	RBU Gap
1 (1–400)	400	57.8	15.0	0.333
2 (401–800)	400	65.5	23.8	0.173
3 (801–1200)	400	68.3	22.8	0.168
4 (1201–1462)	262	67.2	20.2	0.199
Overall	**1,462**	**64.4**	**20.5**	**0.220**

F GEPA-MED Detailed Effectiveness

[TODO: Replace with measured numbers once GEPA-MED experiments complete.]

G Batch-Level Evaluation Stability

To assess evaluation stability across the corpus, we report results stratified by processing batch in Table 8. Skills were processed in sequential batches of 400 (with the final batch containing the remaining 262 skills).

Batch 1 exhibits a lower pass rate (57.8%) and higher RBU gap (0.333) than subsequent batches. This difference is attributable to minor judge prompt refinements applied after Batch 1 processing (within the v3 protocol), which improved consistency in borderline cases. Batches 2–4 show stable performance, with pass rates in the 65.5–68.3% range and RBU gaps between 0.168 and 0.199, confirming that the evaluation framework yields consistent results once calibrated.